

Assignment Part 1: Problem Definition and Data Preparation

1. Clinical Question Definition

The objective of this project is to develop a predictive model for **in-hospital mortality** among adult patients within the **first 24 hours** of their admission to the Intensive Care Unit (ICU). Early risk stratification is critical for clinical decision-making, allowing for prioritized resource allocation and intensified monitoring at a pivotal point where medical interventions can most significantly alter the patient's clinical trajectory.

2. Cohort Selection (Inclusion & Exclusion Criteria)

The study cohort is derived from the WiDS Datathon 2020 dataset. We applied the following criteria to ensure clinical and statistical validity:

- **Inclusion Criteria:** All adult patients (age ≥ 18) with a confirmed ICU admission record.
- **Exclusion Criteria:**
 - **Pediatric Patients (< 18 years):** Excluded due to fundamental physiological differences (e.g., normal vital sign ranges) and different etiologies of critical illness compared to adults.
 - **Readmissions:** Only the first ICU stay per patient was included to maintain independence of observations and prevent data leakage from previous clinical history.
 - **Length of Stay < 4 Hours:** These cases were excluded as they often represent immediate mortality or rapid transfer, where a 24-hour prediction window is less clinically applicable.

3. Feature Selection and Clinical Rationale

Features were selected based on their availability during the first 24 hours and their established clinical relevance:

- **Demographics:** Age and BMI. Age is the most robust independent predictor of physiological reserve and mortality.
- **Physiological Extremes (Vitals):** Minimum and maximum values for Heart Rate, Mean Arterial Pressure (MAP), and SpO2. These capture the patient's hemodynamic stability and acute stress response.
- **Laboratory Data:** Serum Lactate (a marker of tissue perfusion and sepsis), Creatinine (kidney function), and Glucose.
- **Comorbidities:** Pre-existing conditions such as Immunosuppression and Hepatic Failure, which significantly impact the patient's ability to recover from acute insults.

4. Missing Data Strategy

Missingness in EHR data is rarely random (MNAR/MAR). Our strategy treats missingness as a potential clinical signal:

- **Missing Indicators:** For variables with high missingness (e.g., Lactate, Albumin), we created binary flags. In a clinical context, the absence of a test often indicates a clinician's judgment that the patient is hemodynamically stable.
- **Median Imputation:** Used for variables with low missingness (<5%) to preserve sample size without introducing significant bias.
- **Categorical Imputation:** Missing comorbidities were treated as "0" (absence of condition), following standard medical documentation logic where only positive findings are typically recorded.

5. Preliminary Descriptive Analysis (Table 1)

A comparative analysis between "Survivors" and "Non-survivors" in the dataset revealed significant physiological and demographic differences:

- **Age:** Non-survivors were notably older (**Median age: 72 years**) compared to survivors (**Median age: 64 years**).
- **Severity Scores (Baseline):** The baseline APACHE III risk of death probability was significantly higher for non-survivors (**Median probability: 0.16**) compared to survivors (**Median probability: 0.02**). We will use the APACHE III score as our **baseline benchmark** to evaluate the incremental value of our ML model.
- **Biomarkers:** Preliminary analysis shows that non-survivors presented with higher heart rates and lower minimum MAP during the first 24 hours.

Statistical significance was assessed using the **Mann-Whitney U test** for continuous variables (due to non-normal distribution) and the **Chi-squared test** for categorical variables ($p < 0.001$ for age and severity scores), confirming the strong discriminative potential of these features for the modeling phase.